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$$\int e^{\sin(x)}\cos^3(x)dx$$
$$= -\int e^{\sin(x)}\cos(x)(\sin^2(x)-1)dx$$
$$\begin{cases} u = \sin(x) \\ du = \cos(x)dx \end{cases}$$
$$= -\int e^u(u^2-1)du$$
$$\begin{cases} v = u^2-1 \\ dw = e^u du \end{cases} \Rightarrow \begin{cases} dv = 2udu \\ w = e^u \end{cases}$$
$$= -\left(e^u(u^2-1) - 2\int ue^u du\right)$$
$$= -e^u(u^2-1) + 2\left(ue^u - \int e^u du\right)$$
$$\begin{cases} t = u \\ dw = e^u du \end{cases} \Rightarrow \begin{cases} dt = du \\ w = e^u \end{cases}$$
$$= -e^u(u^2-1) + 2ue^u - 2e^u + c$$
$$= -e^u u^2 \cancel{+e^u} + 2ue^u - \cancel{2}e^u + c$$
$$= -\sin^2(x)e^{\sin(x)} + 2\sin(x)e^{\sin(x)} - e^{\sin(x)} + c$$

References